

Enriching continuous Lagrange finite element approximation spaces using neural networks

Michel Duprez¹, Emmanuel Franck², **Frédérique Lecourtier¹** and Vanessa Lleras³

¹Project-Team MIMESIS, Inria, Strasbourg, France

²Project-Team MACARON, Inria, Strasbourg, France

³IMAG, University of Montpellier, Montpellier, France

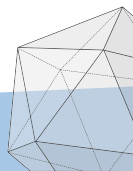
Joint work with:

H. Barucq, F. Faucher, N. Victorion and V. Michel-Dansac.

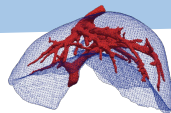
PhD student seminar

IMAG

INSTITUT MONTPELLIERAIN
ALEXANDER GROTHENDIECK



Scientific context



Context : Create real-time digital twins of an organ (e.g. liver).

Objective : Develop an hybrid finite element / neural network method.
accurate quick + parameterized

Parametric elliptic convection/diffusion PDE : For one or several $\mu \in \mathcal{M}$, find $u : \Omega \rightarrow \mathbb{R}$ such that

$$\mathcal{L}(u; \mathbf{x}, \mu) = f(\mathbf{x}, \mu), \quad (\mathcal{P})$$

where $\mathcal{L}$ is the parametric differential operator defined by

$$\mathcal{L}(\cdot; \mathbf{x}, \mu) : u \mapsto R(\mathbf{x}, \mu)u + C(\mu) \cdot \nabla u - \frac{1}{\text{Pe}} \nabla \cdot (D(\mathbf{x}, \mu) \nabla u),$$

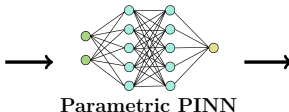
and some Dirichlet, Neumann or Robin BC (which can also depend on μ).

Pipeline of the Enriched FEM

OFFLINE : PINN training

Inputs

$\mathbf{x} \in \Omega$: space coordinates
 $\boldsymbol{\mu} \in \mathcal{M}$: parameter



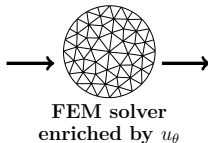
Output

$u_{\theta}(\mathbf{x}, \boldsymbol{\mu})$: prediction of the PDE solution $u(\mathbf{x}, \boldsymbol{\mu})$

ONLINE (fixed $\boldsymbol{\mu}$) : PINN evaluation + Enriched FEM resolution

PINN evaluation

$u_{\theta}(\cdot, \boldsymbol{\mu})$: prediction for the given parameter $\boldsymbol{\mu}$



Output

enriched FEM solution
(depending on the mesh size h)

Complete ONLINE process : quick + accurate

Physics-Informed Neural Networks

Standard PINNs¹ (Weak BC) : Find the optimal weights θ^* , such that

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \left(\omega_r J_r(\theta) + \omega_b J_b(\theta) \right), \quad (\mathcal{P}_\theta)$$

with

residual loss

$$J_r(\theta) = \int_{\mathcal{M}} \int_{\Omega} |\mathcal{L}(u_\theta(\mathbf{x}, \boldsymbol{\mu}); \mathbf{x}, \boldsymbol{\mu}) - f(\mathbf{x}, \boldsymbol{\mu})|^2 d\mathbf{x} d\boldsymbol{\mu},$$

boundary loss

$$J_b(\theta) = \int_{\mathcal{M}} \int_{\partial\Omega} |u_\theta(\mathbf{x}, \boldsymbol{\mu}) - g(\mathbf{x}, \boldsymbol{\mu})|^2 d\mathbf{x} d\boldsymbol{\mu},$$

where u_θ is a neural network, $g = 0$ is the Dirichlet BC.

In $(\mathcal{P}_\theta)$, ω_r and ω_b are some weights.

Monte-Carlo method : Discretize the cost functions by random process.

¹[?]

Conclusion

TODO

Preprint (linear)



References

Preprint (linear)





Appendix 1 : Sec1



A1 – Appendix 1

TODO